
S/4HANA Clean Core Modernization

An engineering blueprint for automated legacy ABAP refactoring via SAP BTP CAP (Node.js/TS) and Developer Extensibility (ABAP Cloud RAP).

01. The Legacy Modernization Bottleneck

Custom legacy code—specifically standard modifications, unreleased database direct table reads, and tightly coupled ABAP logic—represents the single largest barrier to upgrade stability, scalability, and cloud transition in SAP environments.

The SAP **Clean Core** paradigm demands the absolute separation of custom extensions from the standard ERP software. Clean-Core.io acts as a Do-It-Yourself (DIY) automated migration co-pilot, helping architects transition unreleased custom Z-structures into standard-compliant, decoupled cloud-ready architectures.

02. The Dual-Track Extensibility Paradigm



To maintain upgrade stability, custom logic must be refactored into one of two strategic tracks defined by standard SAP extensibility guidelines:

TECHNICAL DIMENSION	IN-APP ABAP CLOUD (RAP)	SIDE-BY-SIDE (BTP CAP)
Core Target	S/4HANA Core Developer Extensibility	SAP BTP (Node.js/TypeScript)
Database Proximity	High (Direct CDS views, synchronous joins)	Decoupled (Dedicated cloud SQL schema)
LUW Boundary	Runs natively inside core transactional block	Asynchronous, decoupled via API/Event Mesh
UI Paradigm	SAP Fiori Elements (strict templates)	React/Angular/Flexible Portal options
Best Fit Case	Real-time business validations and core locks	Burst-scale apps, SaaS, external vendor portals

03. Decoupling the Database Core

Direct database select queries on unreleased tables (such as `BSEG`, `KNA1`, `MARA`) trigger compilation blocks during S/4HANA Public Cloud upgrade cycles. Decoupling requires mapping these entities to released standard public APIs.

The Clean-Core.io transformation pipeline automatically identifies unreleased legacy dependencies and maps them to standard OData/REST scenarios or released CDS views:

LEGACY OBJECT REPLACEMENT

- ✓ `KNA1` → mapped to `I_Customer` (CDS View)
- ✓ `BSEG` → mapped to `I_JournalEntry` (CDS View)
- ✓ `MARA` → mapped to `API_PRODUCT_SRV` (OData)
- ✓ `VBAK` → mapped to `I_SalesOrderTP` (RAP)

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ASYNCHRONOUS EVENT PUB/SUB

For side-by-side extensions, the platform generates a clean ABAP Cloud class (or BAdI implementation) on the ERP side. This trigger intercepts transactional updates in the core database and publishes the event payload asynchronously to the ****SAP Event Mesh****, ensuring near-zero processing coupling.

04. Quality Engineering & Verification Sandboxes

Every transformation is immediately equipped with a sandbox suite to verify logical behavior against standard APIs:

- ✓ **Playwright & TAP Testing Sandbox (CAP):** Auto-generates unit tests and executes them in an isolated server-side sandbox, returning TAP-reporter outputs. Supplying a **BTP Destination JSON** allows testing live connectivity through the ****SAP Cloud Connector**** (utilizing secure Principal Propagation tunnels).
- ✓ **ABAP Unit Test doubles (RAP):** In-app structures are compiled with a full ABAP Unit test suite, verifying method behaviors locally inside the SAP LUW using local mock test doubles.
- ✓ **BPMN 2.0 Architectural Blueprinting:** The L3 process flow is mapped into a standard BPMN 2.0 XML schema with explicit

05. Enterprise Governance & Data Safety

Proprietary legacy source code is a highly confidential business asset. Clean-Core.io is engineered with strict cryptographic and operational safety boundaries to ensure data compliance.

GDPR & EU SOVEREIGNTY INFRASTRUCTURE

All storage and compute layers are hosted entirely within the **Belgium (europa-west1)** GCP region, satisfying European data residency policies (Art. 25 GDPR).

Art. 17 GDPR (Right to Erasure): Triggering account deletion initiates an atomic backend routine that permanently wipes Firebase Auth records, Firestore project documents, and stored analysis artifacts instantly.

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STATELESS PROCESSING & BYOK VAULT

Bring Your Own Key (BYOK): Keys are vaulted inside the user's browser using AES-GCM 256-bit encryption (PBKDF2 with 100,000 iterations). Plaintext keys are never stored on our servers.

Stateless Vertex AI Proxying: All prompts are routed using enterprise-grade stateless configurations. In accordance with Vertex Cloud terms, code is strictly excluded from LLM reinforcement training loops.

06. Executive ROI & Business Value Drivers

Refactoring legacy ABAP code generates immediate business return:

- ✓ **Reduced Regression Testing Cost:** Eliminating core database modifications reduces the average scope of automated and manual regression tests during S/4HANA upgrade cycles by ****~40%****.
- ✓ **Upgrade-Downtime Elimination:** Keeping the core clean guarantees that cloud tenant upgrades execute seamlessly, completely avoiding upgrade-related custom code breakages.
- ✓ **Open-Source Freedom:** Decoupling to SAP BTP CAP allows